

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 15-301-2
COMPRESSION TYPE CONNECTORS For CABLES UP TO 24 kV	Date: 13-03-2025

EDMS 15-301-2

SPECIFICATION
FOR
COMPRESSION TYPE CONNECTORS
For CABLES UP TO 24 kV
(ALUMINUM / COPPER)

Issue: March-2025 / Rev- 2

This revision contains option items that may be requested
by distribution company before bidding

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1. SCOPE

This specification describes the minimum technical requirements for the design, engineering, manufacture, testing, supply and delivery of compression type sleeve connectors for connecting cables/conductors of up to 24 kV. The required connectors shall be suitable for use in both indoor and outdoor installations in the distribution system of Egyptian Electricity Distribution Companies (EDCs).

2. APPLICABLE CODES and STANDARDS

This specification shall be read in conjunction with the latest revisions of the following EDMS specifications:

EDMS 03-300	L.V Cables, Aluminum or copper 0.6/1 KV.
EDMS 03-400	M.V Cables 12/20 KV and 18/30 KV

The equipment/material covered in this specification shall comply with the latest edition/amendment of the following codes and standards. Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply.

IEC 60228	Conductors for Insulated Cables.
IEC 61238-1	Compression and Mechanical Connectors for Power Cables for Rated Voltages up to 30 kV (Um= 36 kV) Part: 1–Test Methods and Requirements.
DIN EN 13600	Copper and copper alloys - Seamless copper tubes for electrical purposes.
DIN EN 755	Aluminum and aluminum alloys - Extruded rod/bar, tube and profiles.
DIN 46267-1	Non tension-proof compression joints for copper conductors.
DIN 46267-2	Non tension-proof compression joints for aluminum conductors.
ISO 2859	Sampling procedures for inspection by attributes.
ASTM B545	Standard Specification for Electrodeposited Coatings of Tin.
NEMA/ANSI CC1	Electric Power Connection for Substations
NEMA/ANSI C119.4	Electric Connectors–Connectors for use between aluminum-to-aluminum, aluminum-to-copper and copper-to-copper conductors.

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3. ENVIRONMENTAL CONDITIONS

The connectors is required to have a good electrical and mechanical performance under the following local service conditions:

Ambient temperature	-5°C to +45°C (50°C is an option according to --EDC requirements),
Maximum relative humidity	95 %
Altitude	Up to 1000 m above mean sea-level
Soil conditions	Exposed to sea water / sewage water (option)

4. DESIGN and CONSTRUCTION REQUIREMENTS

4.1.General:

- All connectors intended for electricity distribution or industrial networks shall be Class-A in which they can be subjected to short-circuit of relatively high intensity and duration.
- All connectors shall meet or exceed the performance of conductor in all respects.
- Manufacturer's drawings shall show the outline of the connectors together with all pertinent dimensions. Manufacturing tolerances shall be as per DIN 755-7 and DIN 13600 for Aluminum and Copper connectors respectively.
- The maximum allowable deviations in inner and outer diameter at any point is ± 0.6 mm from the specified diameter.
- The tolerances in lengths as per applicable DIN standard.
- The maximum allowable decrease in weight is 5% of the specified weight.
- The recommended die is the Hexagonal type, with hudrolic crimping tool.
- The crimping area of the connector should not be less than 50% of the connector total length

4.2.Materials:

- Connectors shall be made entirely from copper (Copper sleeve connectors) or entirely from aluminum (Aluminum sleeve connectors) as specified hereafter according to the request of --EDC.

4.2.1. Copper connectors:

- All copper connectors shall be made of electrolytic tough pitch copper (Cu ETP) of purity not less than 99.9% as per EN 13600. The minimum acceptable conductivity of copper connectors is 96.6% IACS. Copper connectors shall be coated with tin plated surface of thickness 5 μ m to protect against corrosion in accordance with DIN 46267-1.

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4.2.2. Aluminum connectors:

- All aluminum connectors shall be made of seamless extruded seamless tubes of alloy EN-AW-1050A of purity not less than 99.5% in accordance with DIN EN 755-7, DIN 46267-2. The minimum acceptable conductivity of aluminum connectors is 58 % according to IACS.
- The construction details and dimensions are given in the attached drawings and tables, which form part of this specification.
- All connectors shall have smooth bright surface without any form of nicks, deformations, and sharp edges.
- Any aluminum tube of the delivered connectors shall be filled with joint compound (Contact grease with corundum – nonferrous heavy metal inhibitor). The joint compound shall maintain its characteristics up to 90 °C.
- The aluminum tubes of the delivered connectors shall be capped with an easy to be removed plastic plug.

5. TESTING

- Compression connectors shall be tested in accordance to the latest relevant standards and this specification. It shall be the responsibility of the manufacture to perform or to have performed all the tests specified.

5.1 Type Tests

- Type tests had to be performed by accredited laboratories. Single connector of each type Shall be type-tested to cover its lower sizes. The type test reports shall include:
 - a. Short-circuit and heat cycling tests (IEC 61238-1)
 - b. Chemical composition of copper and aluminum (DIN 13600 and DIN 755)
 - c. Tensile tests IEC 61238-1.
- The connectors shall be able to pass the tensile tests prescribed in IEC 61238-1 and the tensile force for each category shall be as follows :

Class	Conductor Material	Nominal cross-sectional area A (mm ²)	Tensile force (N)
Class 1	Aluminum	≤ 500	40 × A
		≥ 630	20000
	Copper	≤ 300	60 × A
		≥ 400	20000

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5.1 Routine Tests

- Routine tests shall be carried out on the charge of the contractor in the presence of EDC representative at the factory. The routine tests shall include:
 - a. Conductivity test to verify the requirement of clause 4.2 of this specification.
 - b. Uniformity of the final product with the sample previously attached to the offer.
 - c. Measurements of dimensions and weights to ensure the compliance with the attached drawings and tables.
- **Sample size:**
 - Test specimens shall be selected at random from each inspection lot in accordance to ISO 2859-1. The number of selected samples shall be as follows:

Lot Size	Sample Size	Lot Size	Sample Size
100 pcs. or less	2 pcs.	1001 to 5000 pcs.	25 pcs.
101 to 500 pcs.	6 pcs.	5001 to 20000 pcs.	40 pcs.
501 to 1000 pcs.	12 pcs.	20001 pcs. to over	60 pcs.

- In case of failure of test sample to comply with routine tests, the EDC reserves the right to disqualify the contractor or postpone the delivery until succeeding all the tests. In this case, any delay of supplying shall be on the charge and responsibility of the contractor.

6. MARKING

Connector identification shall be legibly engraved or indelibly printed included onto both parts of connector surface the following data

- Manufacturer name (shall be marked to where it will not be affected or erased by crimping),
- Conductor size, & die size,
- Number of crimping, crimping positions (narrow die and wide die) shall be clearly indicated on the outside surface in accordance with the attached tables.
- symbol of conductor materials (AL/CU)

7. PACKING

Packing and delivery shall generally be in suitable carton boxes of maximum 15 kg for each. Each box shall bear the following information legibly printed in a visible place:

- Purchase order number / Tender number
- Connector catalogue number
- Manufacturer's name

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- Year of manufacture
- Quantity of pcs. in the box
- Gross weight in kilograms (kg)
- Position of slinging points and other relevant handling instructions.

8. GUARANTEE

- The supplier shall guarantee the products against all defects arising out of faulty design or manufacturing defects or defective material for a period of two (2) years from the date of delivery.
- The supplier shall guarantee the uniformity of the products delivered with the approved samples.

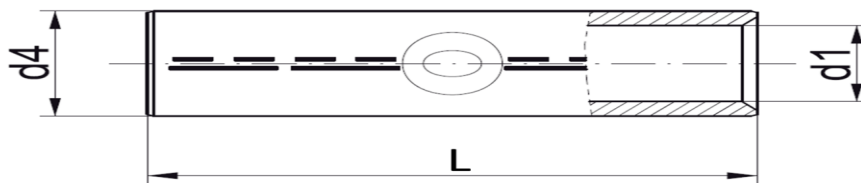
9. SUBMITTALS

- Technical data and original catalogues shall be submitted. The submitted data shall include manufacturer CAD drawings for each of the items offered.
- One specimen of each offered type shall be submitted with the offer for evaluation prior to issuance of the purchase order. Dimensional verification, indelible markings and finishing shall be checked.
- The tenderer must fill in thoroughly the attached technical data schedule for each offered item. Any offer does not accompanied with clear and complete technical data schedule shall be rejected.
- Product type test reports/certificates.
- Approval certificate issued by Egyptian Electricity Holding Company.
- List of deviations and clauses to which exception is taken (if any).

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10. DRAWINGS

Tubular Aluminum Connector



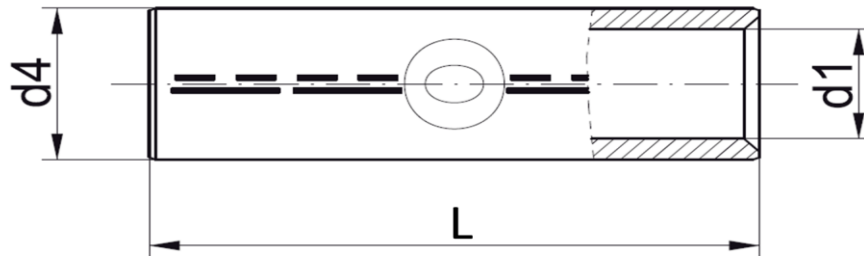
Conductor Cross-section (mm ²)	d1(**)	d4	L(*)	Weight g / pc without inhibitor	No. of Press (crimps) narr/wide
	All dimensions in mm				
10	5	10	55	9.5	3 + 3 / - + -
16	6.5	12	55	11.7	3 + 3 / - + -
25	6.8	12	70	14.5	4 + 4 / 2 + 2
35	8	14	85	24	5 + 5 / 2 + 2
50	9.8	16	85	27.5	5 + 5 / 2 + 2
70	11.2	18	105	44	6 + 6 / 3 + 3
95	13.2	22	105	66	6 + 6 / 3 + 3
120	14.7	22	105	59	6 + 6 / 3 + 3
150	16.3	25	125	95	6 + 6 / 3 + 3
185	18.3	28	125	116	6 + 6 / 3 + 3
240	21	32	145	175	8 + 8 / 3 + 3
300	23.3	34	145	187.5	8 + 8 / 3 + 3
400	26	38	210	342	- + - / 4 + 4
500	29	44	210	487.5	- + - / 4 + 4
630	32	44	210	403	- + - / 4 + 4
800	36	52	210	626	- + - / 4 + 4
1000	41.5	58	210	730	- + - / 4 + 4

(*) :For M.V joints connector length “L” should be based on manufacturer design.

(**) For M.V joints of cables size 150, 240, 300, and 400mm² inner diameter “d1” accepted to reduce by 1mm, in-additional of applying tolerances in AM table.

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Tubular Copper Connector



Conductor Cross-section (mm²)	d1	d4	L(*)	Weight g / pcs	No. of Press (crimps) narr/wide
	All dimensions in mm				
10	4.5	6	30	33.1	1 + 1 / - + -
16	5.5	8.5	50	14.5	2 + 2 / - + -
25	7	10	50	17.6	2 + 2 / - + -
35	8.2	12.5	50	30.7	2 + 2 / - + -
50	10	14.5	56	42	3 + 3 / - + -
70	11.5	16.5	56	54.6	3 + 3 / - + -
95	13.5	19	70	87	4 + 4 / 2 + 2
120	15.5	21	70	98.3	4 + 4 / 2 + 2
150	17	23.5	80	145	4 + 4 / 2 + 2
185	19	25.5	85	168.5	5 + 5 / 2 + 2
240	21.5	29	90	236	- + - / 2 + 2
300	24.5	32	100	294	- + - / 2 + 2
400	27.5	38.5	150	759	- + - / 3 + 3
500	31	42	160	890	- + - / 3 + 3
630	34.5	44	160	834	- + - / 3 + 3
800	38.4	52	200	1544	- + - / 5 + 5
1000	44	58	200	1995	- + - / 5 + 5

(*) :For M.V joints connector length "L" should be based on manufacturer design.

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11. TECHNICAL DATA SCHEDULE

-- EDC Inquiry No: _____ Item No: _____

No.	DESCRIPTION	-- EDC Specified Values	Vendor Proposed Values
Connectors			
1	Manufacturer	---	
2	Manufacturer Catalogue No.	---	
3	Country of Origin	---	
4	Material	Cu / Al	
5	Material Compliance	Yes / No	
6	Material purity (%)	Cu 99.9 % / Al 99.5 %	
7	Material conductivity (IACS)	Cu 96.6 % / Al 58 %	
8	Cable / Conductor Size (mm ²)	10 up to 1000	
9	Tin-Plating Thickness (µm)	- / 5(for copper only)	
10	Submittals Required Included	Yes / No	

I / We guarantee the technical data given above for the offered material/equipment.

Description	Manufacturer of Material/Equipment	Vendor/Supplier
Vendor/Supplier		
Location and Office Address		
Name and Signature of Authorized Representative with Date		
Official Seal / Stamp		